

California State Polytechnic University, Pomona
Autonomous Vehicle Laboratory
IGVC 2026 Design Report

ROBOT NAME



I hereby certify that the development of the vehicle, *ROBOT NAME*, as described in this report, is equivalent to the work involved in a senior design course.

Submitted May XX, 2026

Software	Mechanical	Electrical
Name	Name (Captain)	Name
Name	Name	Name
Name	Name	Name
Name		

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1. Design Process, Team Identification and Organization

1.1 Introduction

The Autonomous Vehicle Laboratory (AVL), representing California State Polytechnic University, Pomona (Cal Poly Pomona) and the College of Engineering, proudly presents *ROBOT NAME*, our entry for the 2026 Intelligent Ground Vehicle Competition (IGVC) in the AutoNav category. AVL is a multidisciplinary student organization within the College of Engineering, with a dedicated IGVC team of XX students. The team is advised by ADVISOR NAME(S), faculty member(s) at Cal Poly Pomona.

Cal Poly Pomona’s learn-by-doing philosophy is at the core of AVL’s approach. Every component of the vehicle—from mechanical design to embedded firmware to autonomous navigation software—is designed, built, and tested by students. This is the team’s first entry in the IGVC competition.

1.2 Team Organization

The IGVC team is organized into three specialized subteams: electrical, mechanical, and software. A team captain serves as the primary point of contact with AVL leadership and faculty advisors, providing regular updates on progress and coordinating across subteams.

- **Mechanical:** Chassis design, drivetrain, sensor mounting, weatherproofing
- **Electrical:** Power distribution, motor controllers, PCB design, CAN bus, e-stop
- **Software:** Perception, navigation, localization, firmware, GUI

To promote communication and prevent silos, the team holds weekly full-team meetings in addition to subteam work sessions. All design decisions are documented and version-controlled.

1.3 Design Process

To develop *ROBOT NAME*, the team implemented an iterative design process that focuses on a cycle of researching, designing, prototyping, and testing.

1. **Define** — Establish system-level requirements from competition rules.
2. **Research** — Survey existing solutions, reference designs, and prior IGVC reports.
3. **Design** — Create detailed CAD models, schematics, and software architecture.
4. **Prototype** — Build rapid prototypes (3D-printed parts, breadboard circuits, simulation).
5. **Test** — Validate each subsystem independently, then integrate.
6. **Implement** — Finalize manufacturing and deploy to the competition vehicle.

The team began by analyzing design reports from prior IGVC winners and performing a full re-read of the 2026 competition rules to derive a list of rule-based requirements. From these, the team established high-level design goals. With the system-level requirements defined, subteams proceeded to research, design, build, and test the necessary components. As each

subsystem was completed, integration testing ensured that all components worked cohesively and that the robot met its overall requirements.

2. System Architecture

2.1 Mechanical and Electrical Architecture

At a high level, *ROBOT NAME* consists of several independent subsystems: chassis and tracked drivetrain, vision, sensors, and electronics.

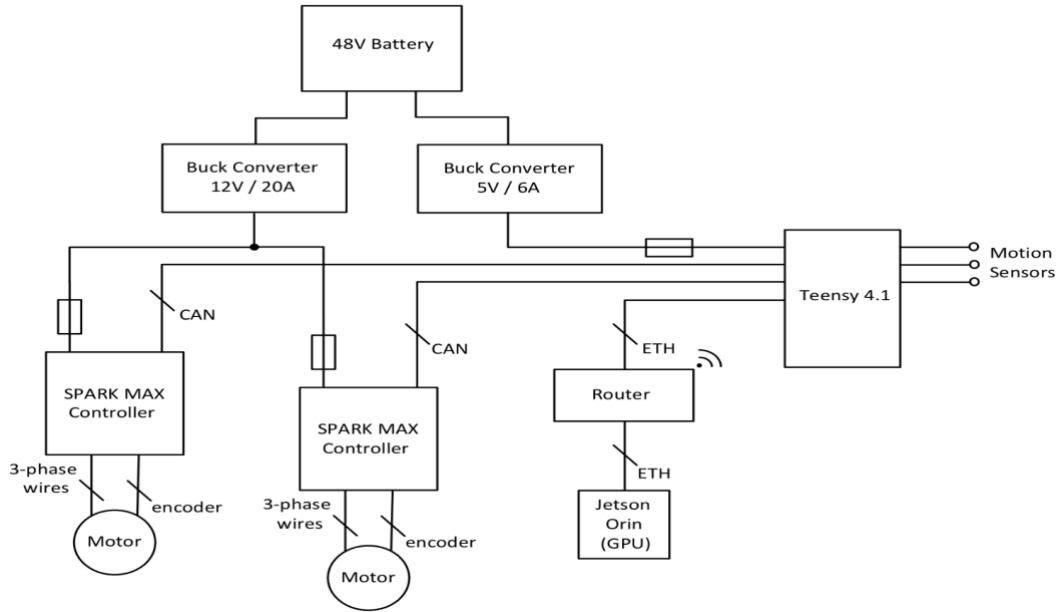


Figure 1: High-level electronic and power system architecture.

Table 1: Major subsystems overview.

Subsystem	Description
Chassis	AndyMark Raptor Track Drive with 80/20 T-slot aluminum extrusion frame and 1/8 in diamond plate steel base (21.5 in $\times$ 30 in)
Drivetrain	Tracked differential drive (skid-steer) with 2 $\times$ REV SparkMAX controllers and NEO brushless motors
Compute	NVIDIA Jetson Orin (GPU) for vision processing and object detection
Microcontroller	Teensy 4.1 — CAN bridge between main computer and motor controllers
Sensors	3 $\times$ Stereolabs ZED X stereo cameras (front-left, front, front-right), Velodyne VLP-16 3D LiDAR, Xsens MTi-680G GNSS/INS
Power	48 V battery with dual buck converters (12 V/20 A for motors, 5 V/6 A for Teensy 4.1)

2.2 Robot Safety

ROBOT NAME was designed with safety as a fundamental principle. Power is delivered to the drive motors only when the following conditions are simultaneously met:

1. The **remote emergency stop** is inactive (wireless e-stop not triggered).
2. The **physical emergency stop** button on the vehicle is not pressed.
3. The **software emergency stop** has not been asserted.

If any one of these is triggered, motor power is immediately cut. To prevent accidental re-engagement, each of the three signals is also monitored redundantly in software. As a result, the robot will not send any non-zero command to the motors unless all safety interlocks are satisfied.

2.3 Software Architecture

The software is split into four major subsystems: sensors, perception, localization, and navigation.

- **Sensors** — Stereo camera capture (2× ZED X), 3D LiDAR point clouds (Velodyne VLP-16), GNSS/INS readings (Xsens MTi-680G), motor encoder feedback
- **Perception** — GPU-accelerated HSV lane detection on stereo camera feeds and LiDAR-based obstacle detection (Jetson Orin)
- **Localization** — Sensor fusion of track odometry, GNSS position, and 9-axis inertial data for position and heading estimation
- **Navigation** — Path planning and motor command generation

Organizing the code in this modular way allows for easy testing of each subsystem independently as well as making testing different algorithms or parameters quick and easy.

2.4 Robot Communication Bus

Data is communicated between devices using the **CAN bus** protocol, a robust standard widely used in the automotive industry. The Teensy 4.1 microcontroller serves as the CAN bridge between the main onboard computer and the REV SparkMAX motor controllers.

The CAN bus operates at **1 Mbit/s** using 29-bit extended frames. A CAN transceiver (SN65HVD230) converts the Teensy’s logic-level signals to differential CAN_H / CAN_L. The bus is terminated with $120\ \Omega$ resistors at each end.

2.5 Bill of Materials

The total cost to construct *ROBOT NAME* was approximately \$X,XXX.

Table 2: Major COTS components.

Major COTS Item	Qty	Market Cost (USD)	Team Cost (USD)
NVIDIA Jetson Orin AGX	1	\$TODO	\$TODO
Stereolabs ZED X stereo camera	3	\$TODO	\$TODO
Velodyne VLP-16 3D LiDAR	1	\$TODO	\$TODO
Xsens MTi-680G GNSS/INS	1	\$TODO	\$TODO
48 V Battery	1	\$TODO	\$TODO
AndyMark Raptor Track Drive	1	\$TODO	\$TODO
REV SparkMAX + NEO Motor	2	\$TODO	\$TODO
Teensy 4.1	1	\$30.00	\$30.00
CAN Transceiver (SN65HVD230)	1	\$5.00	\$5.00
Buck Converters (12 V + 5 V)	2	\$TODO	\$TODO
Router	1	\$TODO	\$TODO

3. Effective Innovations

3.1 Innovation 1: Cost-Effective Tracked Platform with Modular Frame

As first-year participants, the team made the strategic decision to build on a commercially available AndyMark Raptor Track Drive platform combined with 80/20 T-slot aluminum extrusions. This approach provides several advantages over a fully custom chassis:

- **Reduced development time** — By using a proven tracked drive platform, the team could focus development effort on sensors, electronics, and software rather than chassis fabrication.
- **Superior traction** — The tank-track configuration spreads the vehicle's weight over a larger contact area compared to wheels, providing better traction and stability on uneven terrain and the competition ramp.
- **Modular reconfigurability** — The T-slot extrusion frame allows components to be repositioned or swapped without permanent modifications, enabling rapid iteration during development and easy adaptation for future competitions.

3.2 Innovation 2: GPU-Accelerated Vision with Jetson Orin

The team selected the NVIDIA Jetson Orin as the main onboard computer, leveraging its GPU capabilities for computationally intensive vision processing and object detection. Unlike CPU-only solutions, the Jetson Orin enables real-time image processing pipelines that would otherwise introduce unacceptable latency. This allows the vehicle to run advanced perception algorithms while maintaining the responsiveness required for safe autonomous navigation.

3.3 Custom CAN Firmware for SparkMAX Control

The team developed custom firmware for the Teensy 4.1 microcontroller that communicates directly with REV SparkMAX motor controllers over CAN bus. This eliminates the need for a roboRIO or additional middleware, reducing cost, latency, and complexity. The firmware handles:

- Non-RIO heartbeat broadcast to keep SparkMAXes enabled
- Percent-output, velocity, and position command modes
- Periodic encoder feedback parsing for odometry
- Serial interface for integration with the main onboard computer

4. Mechanical Design

4.1 Overview

As first-year participants, the team prioritized a simple, modular, and cost-effective mechanical design. This approach allowed the team to concentrate its budget toward sensors and electronics critical for autonomous navigation. The vehicle is built on a commercially available tracked drive platform from AndyMark®, the Raptor Track Drive. This track-based structure provides a reliable and robust base for mobility. For design flexibility, T-slot aluminum extrusions were mounted directly to the chassis, enabling rapid prototyping and testing of different structural configurations throughout the development process.

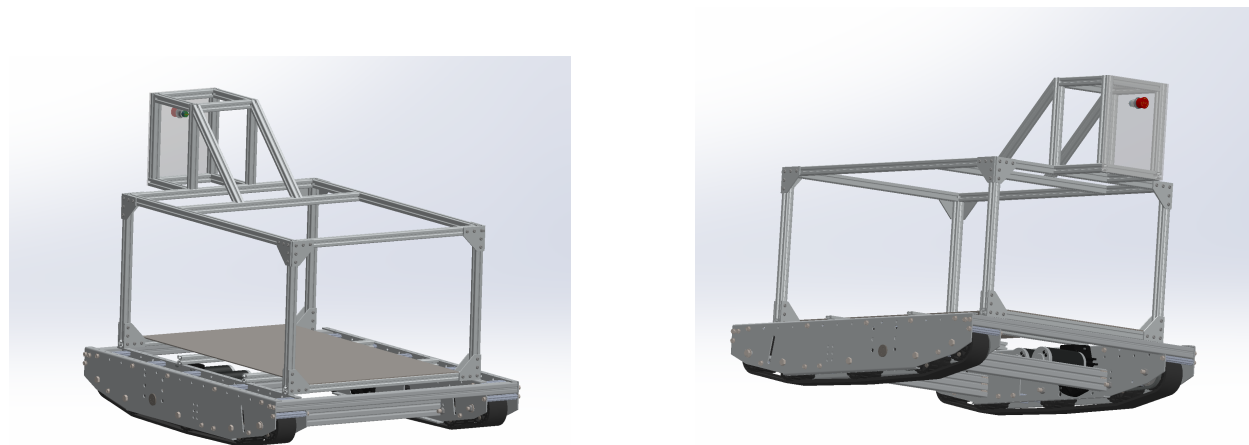


Figure 2: CAD renders of *ROBOT NAME*: front view (left) and rear view (right).

4.2 Drivetrain

ROBOT NAME utilizes a tracked differential drive system powered by two REV NEO brushless motors, each controlled by a REV SparkMAX motor controller. This configuration was chosen for both its mechanical and computational simplicity, as well as traction performance in outdoor environments.

The fixed belt-tracked system allows for skid-steer operation where each motor drives one side of the track independently. Straight motion is achieved when both motors rotate at the same speed in the same direction, while turning is accomplished by varying the relative speeds of each motor:

- **Pivot Turn** — One track moves forward while the other stays still; the robot pivots around the stationary track.
- **In-Place Rotation (Spin)** — The tracks rotate in opposite directions at the same speed ($v_L = -v_R$); the robot rotates around its center point.
- **Gradual Curve** — One track rotates faster than the other ($v_L > v_R$ to turn right).

This approach eliminates the need for four independent motors or steering mechanisms, reducing system complexity in both mechanical design and control software. Using the track

belts instead of regular wheels, the weight of *ROBOT NAME* is spread over a larger area, increasing traction and stability across uneven terrain and obstacles encountered on the course.

Table 3: Drivetrain specifications.

Parameter	Value
Drive type	Tracked differential (skid-steer)
Drive platform	AndyMark Raptor Track Drive
Motors	2× REV NEO brushless
Motor controllers	2× REV SparkMAX
Gear ratio	TODO
Designed top speed	TODO
Track width	TODO

4.3 Frame and Sensor Mast

The structural frame was constructed using 80/20[®] 20-Series T-slot aluminum extrusions (80/20 Inc., USA). These extrusions allow for a modular and reconfigurable design that can be adapted for future competitions. To improve safety during assembly and operation, T-slot nuts were utilized within the extrusion channels, eliminating exposed threaded ends and reducing the risk of sharp edges. Attached to the base chassis is a 1/8 in thick diamond plate steel base (21.5 in × 30 in), offering a rigid and large mounting surface for heavy components.

Weight distribution was a prioritized design consideration. The battery pack was mounted toward the rear of the chassis to act as a counterbalance to the 20 lb payload, which was positioned toward the front of the vehicle on a 3D-printed mount. Electronics were centrally located and mounted on 3D-printed brackets attached to a carbon fiber plate, preventing direct contact with the steel base and protecting sensors and components from electrical interference.

A sensor mast is mounted at the center/front of the robot to elevate the three ZED X stereo cameras, the Velodyne VLP-16 3D LiDAR, and the Xsens MTi-680G GNSS antenna. The ZED X cameras are mounted as an angled trio—one facing forward and two angled outward to the front-left and front-right—to maximize lateral coverage of lane lines on both sides of the vehicle while preserving a dedicated forward view for long-range lane tracking. The Velodyne VLP-16 is mounted above the cameras to preserve an unobstructed 360° horizontal field of view for obstacle detection. Elevating the cameras provides a wider field of view and allows the image to be projected onto a 2D plane with minimal distortion, while elevating the GNSS antenna reduces multipath interference.

4.4 Weatherproofing

ROBOT NAME utilizes acrylic panel walls integrated into the T-slot frame for protection from rain and sun. The acrylic sheets are cut-to-fit inside the T-slot extrusion grooves, with

weatherstripping gaskets filling the gaps to ensure a secure fit that remains in place even with vibration during operation. The top panel is angled for water runoff and includes a sliding door for maintenance, wiring, and payload handling. All cable pass-throughs are sealed to prevent water ingress.

4.5 Constraints

To satisfy the competition size requirements, *ROBOT NAME* was extended through the addition of an 8×6 in rear-mounted enclosure for the mechanical E-stop button, extending the length of the vehicle over the minimum 3 ft required. This structure was mounted at 2 ft from the ground, and the E-stop button was centrally positioned for easy access.

5. Electronic and Power Design

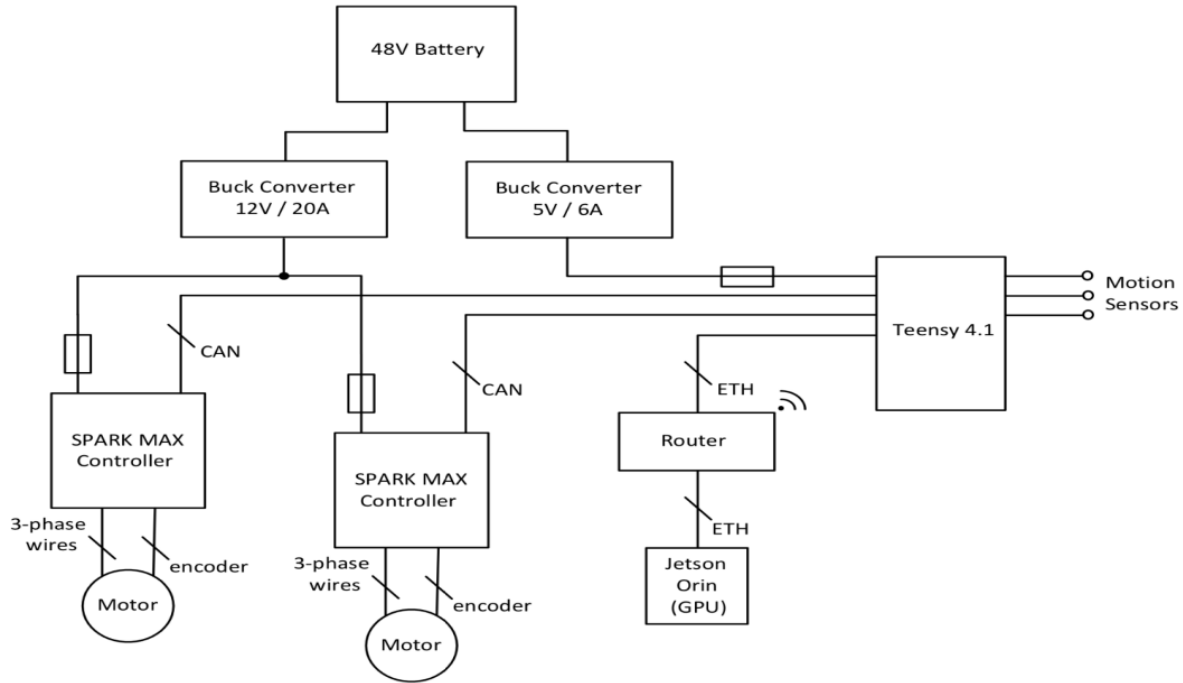


Figure 3: Electronic and power architecture of *ROBOT NAME*.

5.1 Overview

The electronic system of *ROBOT NAME* is designed for modularity and repairability. The architecture features the following key components:

- **NVIDIA Jetson Orin AGX** — Main onboard GPU computer running HSV lane detection on the ZED X stereo feeds and LiDAR-based obstacle detection
- **Teensy 4.1** — Single microcontroller node acting as the CAN bridge between the Jetson and the REV SparkMAX motor controllers, handling motor commands and encoder feedback
- **CAN transceiver** — SN65HVD230 converting logic-level to differential CAN
- **2× REV SparkMAX** — Brushless motor controllers with built-in PID
- **3× Stereolabs ZED X** — GMSL2 stereo cameras (front-left, front, front-right) for lane detection
- **Velodyne VLP-16** — 16-channel 360° 3D LiDAR for obstacle detection, connected via Ethernet
- **Xsens MTi-680G** — GNSS/INS unit providing GPS fixes and 9-axis inertial data (RTK disabled per IGVC rules)
- **Router** — Central network hub connecting Jetson Orin, Velodyne VLP-16, and Teensy 4.1 via Ethernet, with Wi-Fi to remote laptop
- **E-stop system** — Hardware relay controlled by physical + wireless + software interlocks

- **Safety lights** — RGB indicator lights driven from the Jetson

5.2 Power Distribution

The power system consists of a 48 V battery wired to two separate buck converters. The first buck converter brings the voltage down to 12 V with a maximum output of 20 A, which provides sufficient current to the 12 V motors via the SparkMAX controllers. The second buck converter brings the voltage down to 5 V with a maximum output of 6 A, which is used to power the Teensy 4.1 microcontroller. Both buck converters run through multiple fuses before reaching their respective devices, providing overcurrent protection throughout the power distribution system.

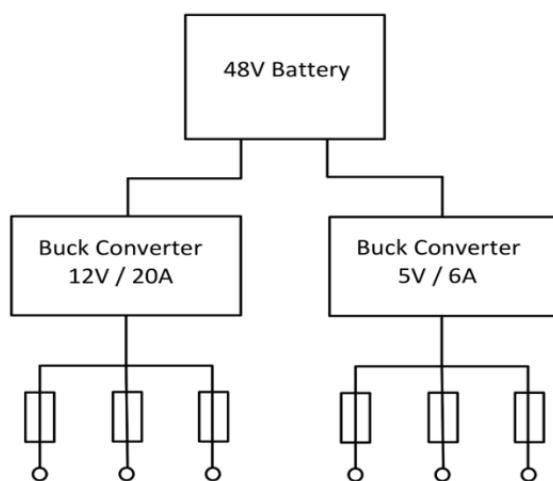


Figure 4: Power distribution topology: 48 V battery to dual buck converters with fused outputs.

Table 4: Power budget.

Component	Voltage (V)	Peak Current (A)	Avg Current (A)
Jetson Orin AGX	19 (via buck)	4.0	2.0
Teensy 4.1	5 (via 5 V/6 A buck)	0.1	0.05
SparkMAX + motors	12 (via 12 V/20 A buck)	TODO	TODO
3× ZED X cameras	12	2.3	1.8
Velodyne VLP-16	12	0.8	0.6
Xsens MTi-680G	5 (USB)	0.2	0.15
Router	12	0.8	0.5
Safety lights	12	0.5	0.3
Total	—	TODO	TODO

5.3 Computer Network

The computer network is centered around a router that connects both the NVIDIA Jetson Orin and the Teensy 4.1 microcontroller via Ethernet. The Jetson Orin is used for computationally intensive tasks such as vision processing and object detection, while the Teensy 4.1 handles system data including motor control and sensor feedback over CAN bus. The router also transmits system data to a remote laptop via Wi-Fi, enabling real-time monitoring and debugging during operation.

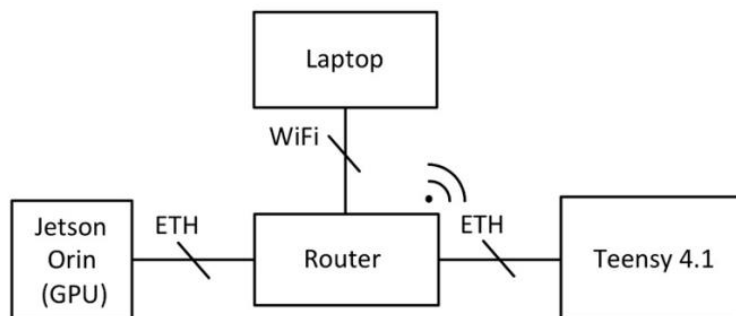


Figure 5: Computer network topology.

5.4 Motor Control

ROBOT NAME controls its two drive motors by commanding the SparkMAX controllers over a dedicated CAN bus. The Teensy 4.1 serves as the CAN bridge, translating high-level commands from the main computer (received over Ethernet via the onboard router) into SparkMAX CAN frames. Using the data received from the SparkMAX controllers, the system implements PID control to maintain correct speeds and position.

The SparkMAX controllers use REV Robotics’ proprietary CAN protocol with 29-bit extended identifiers. The CAN ID is structured as follows:

Table 5: SparkMAX CAN ID bit-field layout (29-bit extended).

Field	Bits	Width	Value
Device Type	28:24	5 bits	2 (Motor Controller)
Manufacturer	23:16	8 bits	5 (REV Robotics)
API Class	15:10	6 bits	Varies by command
API Index	9:6	4 bits	Varies by command
Device ID	5:0	6 bits	1–62 (unique per controller)

Key API commands used:

- **Non-RIO Heartbeat** (Class 11, Index 2) — Sent every 20 ms to keep controllers enabled
- **Percent Output** (Class 0, Index 2) — Motor duty cycle from -1.0 to $+1.0$

- **Velocity Setpoint** (Class 1, Index 2) — Target RPM using SparkMAX internal PID
- **Encoder Feedback** (Class 6, Index 2) — Periodic drive encoder position reports

All floating-point values are encoded as IEEE 754 single-precision floats in little-endian byte order. By offloading the PID control loop onto the SparkMAX controllers, we simplify the firmware and reduce end-to-end latency.

5.5 Safety Lights

Safety lights on the vehicle indicate the autonomous state as required by IGVC rules.

Table 6: Safety light color mapping.

Light Color	Meaning
White	Standard operation
Yellow	Waypoint following activated
Green	Waypoint reached
Red	Unable to move forward / error

5.6 Emergency Stop and Safety

Safety is a top priority for the vehicle and is at the core of our electrical design. The emergency stop circuit features three interlocks that must all be disengaged for the motors to receive power:

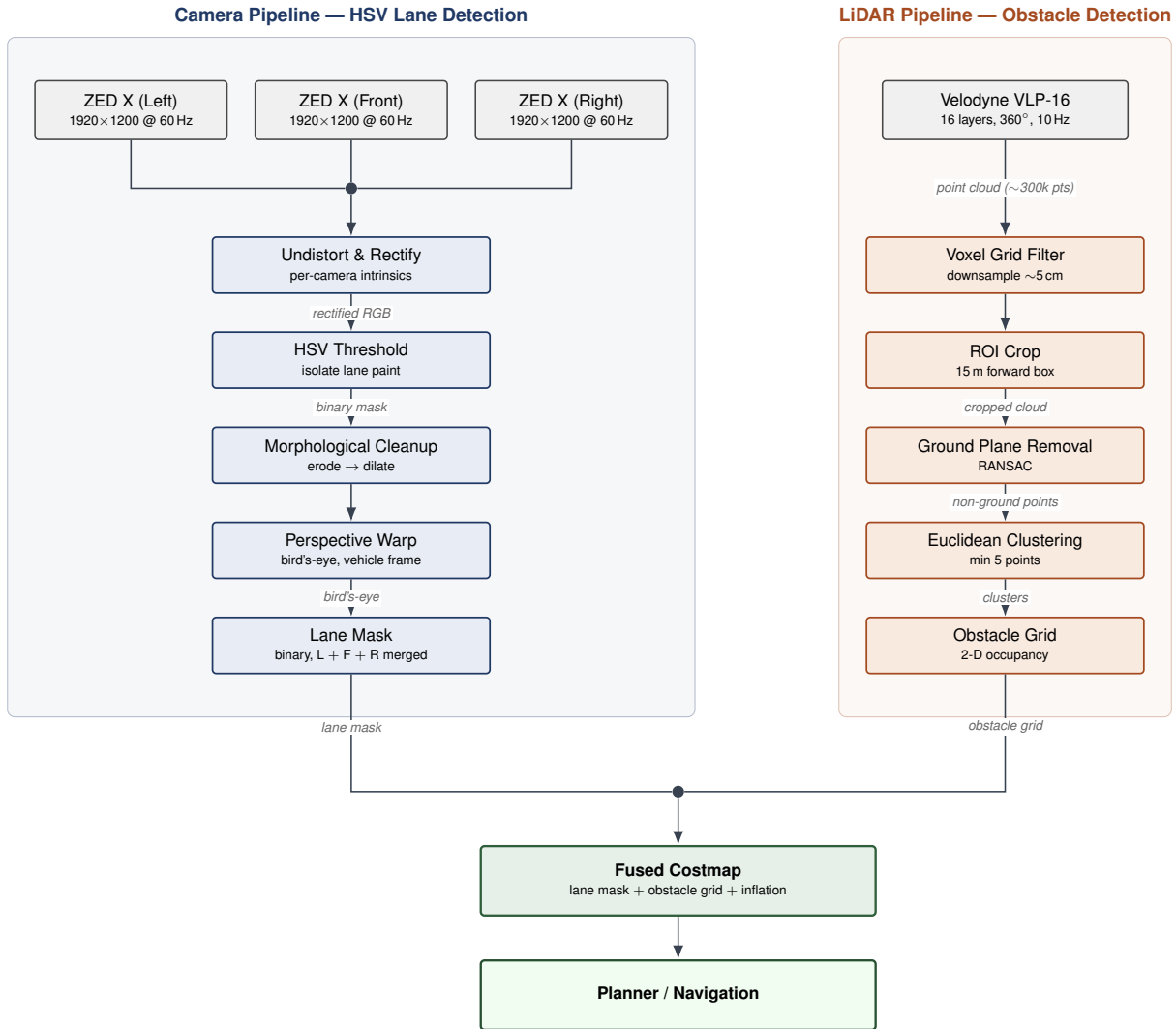
1. **On-vehicle button** — Physical red mushroom-head e-stop button.
2. **Remote button** — Wireless e-stop with >100 ft range, as required by IGVC rules.
3. **Software interlock** — Software-controlled relay that is only enabled when all nodes report healthy status.

Motor power is gated by a solid-state relay controlled directly by the e-stop line. If any interlock is triggered, power to the motor controllers is immediately cut regardless of software state.

6. Software Systems

6.1 Overview

Our software uses the **Robot Operating System (ROS 2)**, a robotics middleware that is widely used in both academia and industry. It is based on a publisher-subscriber model, with individual processes called nodes sending messages to each other. This allows for much of the software work to be developed in parallel in separate packages.



All perception stages execute on the NVIDIA Jetson Orin AGX.

Figure 6: Perception pipeline. Two parallel branches process data from the three ZED X stereo cameras (HSV lane detection) and the Velodyne VLP-16 LiDAR (3D obstacle detection), merging into a unified costmap consumed by the planner.

6.2 Sensor Suite

ROBOT NAME uses a complementary sensor suite that splits perception responsibilities between cameras and LiDAR: the cameras find painted lane lines through HSV color segmentation, and the LiDAR detects physical obstacles through 3D point cloud processing. This split avoids relying on any single modality for safety-critical obstacle detection.

Table 7: Sensor suite specifications.

Sensor	Model	Key Specs	Function
Stereo camera (L)	Stereolabs ZED X	1920×1200 stereo RGB, 60 fps, GMSL2, wide FOV	Front-left lane detection
Stereo camera (F)	Stereolabs ZED X	1920×1200 stereo RGB, 60 fps, GMSL2, wide FOV	Front lane detection
Stereo camera (R)	Stereolabs ZED X	1920×1200 stereo RGB, 60 fps, GMSL2, wide FOV	Front-right lane detection
3D LiDAR	Velodyne VLP-16	16 layers; 360°; 100 m range; 10 Hz	Obstacle detection
GNSS/INS	Xsens MTi-680G	9-axis; 100 Hz; ±2–5 m (RTK disabled)	Localization
Compute	NVIDIA Jetson Orin AGX	GPU; Ethernet; GMSL2 capture	Perception

Per IGVC rules, RTK corrections are not permitted, so the Xsens MTi-680G operates in standalone GNSS/INS mode. This yields approximately $\pm 2\text{--}5\text{ m}$ position accuracy; the localization stack compensates by fusing GNSS with wheel/track odometry and the unit's 9-axis inertial data for short-horizon dead reckoning between waypoints.

6.3 Lane Detection via HSV Computer Vision

ROBOT NAME uses a classical computer vision approach on the three ZED X stereo cameras to find painted course lines and project them into a drivable-area map. The front-left, front, and front-right ZED X cameras are processed independently and then merged in the vehicle frame. This three-camera arrangement provides overlapping coverage of lane lines directly in front of the robot and on both sides, which is more reliable than a single forward camera when the vehicle is turning or when a lane line falls outside the forward field of view.

Each camera feed is processed through the following pipeline on the Jetson Orin:

1. **Camera calibration** — Correct lens distortion using the factory ZED X intrinsics and a team-calibrated extrinsic from each camera to the vehicle frame.
2. **Perspective transform** — Project the rectified image onto a top-down 2D ground plane using the known mounting geometry.
3. **HSV thresholding** — Convert to HSV color space and threshold on hue/saturation/-value to isolate white and colored lane paint while rejecting grass and shadows.
4. **Morphological cleanup** — Apply erosion/dilation to suppress speckle and bridge short gaps in the detected line pixels.
5. **Occupancy grid fusion** — Merge the front-left, front, and front-right lane masks into a unified vehicle-frame lane mask for the planner.

An auto-calibration routine samples the ground at startup and in response to large lighting changes to adapt the HSV thresholds to current conditions.

6.4 LiDAR-Based Obstacle Detection

Physical obstacles (barrels, cones, fences, pedestrians) are detected using the Velodyne VLP-16. Point clouds stream to the Jetson Orin over Ethernet at 10 Hz and are processed through the following pipeline:

1. **Point cloud filtering** — Crop to a region of interest around the vehicle and down-sample with a voxel grid.
2. **Ground plane removal** — Estimate and remove the ground plane so that terrain does not register as an obstacle.
3. **Clustering** — Group the remaining non-ground points into discrete obstacle clusters.
4. **Occupancy update** — Project clusters into a 2D occupancy grid aligned with the lane mask from the camera pipeline.

The occupancy grid produced by the LiDAR pipeline and the lane mask produced by the HSV pipeline are combined into a single costmap that the navigation layer consumes for path planning.

6.5 Navigation

TODO: Describe your path planning and navigation algorithm.

The navigation system receives the obstacle map from the perception subsystem and the current position from the localization subsystem. It computes motor commands that guide the vehicle toward the next GPS waypoint while avoiding detected obstacles.

6.6 Localization

Autonomous localization is required for qualifying for and completing the IGVC AutoNav challenge. Waypoint navigation is only possible with accurate and precise location data.

ROBOT NAME uses an extended Kalman filter that fuses the following inputs:

- **Track odometry** — Encoder feedback from the SparkMAX motor controllers, transmitted through the Teensy 4.1 over CAN bus and used for short-horizon dead reckoning.
- **GNSS data** — Xsens MTi-680G standalone GNSS fix at 5 Hz. RTK corrections are not used per IGVC rules, giving approximately $\pm 2\text{--}5\text{ m}$ absolute position accuracy.
- **IMU data** — 9-axis inertial data (3-axis gyro, 3-axis accelerometer, 3-axis magnetometer) from the Xsens MTi-680G at 100 Hz, providing heading and angular velocity.

The Xsens MTi-680G connects directly to the Jetson Orin over USB and runs its internal sensor fusion at 100 Hz before publishing to the localization node. Because competition rules prohibit RTK, the localization stack is tuned to weight odometry and inertial data more heavily than the raw GNSS fix between waypoint captures, using GNSS primarily to bound drift over longer distances.

This fused location data is consumed by the path planner as the robot's current position and is used to determine waypoint arrival and distance from the start of the course.

6.7 Drive Control

The Jetson Orin publishes desired linear and angular velocity commands. These are converted to individual left/right track speeds using the differential drive kinematic model:

$$v_L = v - \frac{\omega \cdot d}{2} \tag{1}$$

$$v_R = v + \frac{\omega \cdot d}{2} \tag{2}$$

where v is the desired linear velocity, ω is the desired angular velocity, and d is the track width (distance between track centers).

The computed track speeds are sent to the Teensy 4.1 over Ethernet (via the onboard router), which translates them into SparkMAX CAN commands. Encoder feedback from both motors is read by the Teensy and sent back to the Jetson Orin for odometry computation.

6.8 Configuration and GUI

TODO: Describe any GUI, configuration system, or parameter tuning tools.

Configuration of tunable parameters is handled via a custom GUI / ROS parameter server running on the main computer. This allows the team to tune the robot on-the-fly without recompiling code.

6.9 State Machine

The robot operates using a state machine that manages transitions between the following system states:

- **Disabled** — Motors off, all subsystems idle.
- **Manual** — Operator control via gamepad/controller for testing.
- **Autonomous** — Full autonomy, running perception + navigation + localization.
- **Emergency Stop** — Immediate motor cutoff, requires manual reset.

Each software node also maintains its own local state (off, warming, ready, operating, error) to ensure all subsystems are healthy before autonomous operation begins.

6.10 Logging

TODO: Describe your data logging strategy for post-run analysis.

After every run, a comprehensive log file is created containing sensor readings (camera feeds, GPS, IMU, encoder feedback), diagnostic metrics, navigation targets, and debug information.

7. Cyber Security Analysis

7.1 NIST RMF Process

Security vulnerabilities in self-driving vehicles pose significant risks. To effectively mitigate these threats, the team follows the NIST Risk Management Framework (RMF), which provides a structured process for identifying, assessing, and managing risks throughout the system's life cycle.

7.2 Vehicle Threats

Table 8: Threat assessment matrix.

Subsystem	Threat Description	Confid.	Integrity	Avail.
GPS Waypoints	GPS waypoint tampering could lead the vehicle to drive in unintended ways.	Low	High	High
Remote E-stop	Malicious interference could disable or trigger the e-stop.	Moderate	High	High
Onboard Computer	Malware or unauthorized access could compromise system control.	High	High	High
Odometry	Sensor data tampering can result in incorrect state estimation.	Low	High	Moderate
CAN Bus	CAN frame injection could send unauthorized motor commands.	Moderate	High	High
Git Repository	Unauthorized access could inject malicious code.	High	High	Moderate

7.3 Cyber Controls and Implementation

Table 9: Security controls.

Subsystem	Control	Implementation
GPS Waypoints	SI-4 System Monitoring	Display GPS waypoints to the user and validate using range checks.
Remote E-stop	AC-3 Access Enforcement	Authenticate and/or encrypt the e-stop radio channel.
Onboard Computer	SC-28 Protection of Information at Rest	Use secure boot, enable SSH authentication, and disable passwords.
Odometry	SI-7 Information Integrity	Validate sensor input using range checks and sensor fusion.
CAN Bus	SC-8 Transmission Confidentiality	Monitor for anomalous CAN traffic; heartbeat watchdog timeout.
Git Repository	CM-5 Access Restrictions for Change	Enforce code access control with role-based permissions.

8. Vehicle Analysis

8.1 Lessons Learned

TODO: Describe the key lessons your team learned during the design and build process.

8.2 Failure Points and Resolutions

Table 10: Failure points and resolutions.

Category	Failure Point	Cause	Resolution
Mechanical	Fasteners loosen	Vibrations from driving	Loctite and Nyloc nuts
Electrical	E-stop connection lost	Remote goes out of range	Heartbeat timeout stops robot
Electrical	CAN bus errors	Wiring / termination issues	Added proper $120\ \Omega$ termination
Software	Obstacle misdetection	Changing weather / lighting	Auto HSV calibration at startup
Software	Sensor data loss	Cable disconnect / node crash	Watchdog monitors node health

8.3 Safety, Reliability, and Durability

ROBOT NAME was designed from the ground up to be safe, reliable, and durable in all conditions. Through dozens of test runs in various scenarios—sensor loss, changing weather conditions, e-stop testing—the team has verified that the robot reliably activates all of its safety measures.

TODO: Describe specific reliability testing you performed.

8.4 Performance Metrics

Table 11: Performance metrics.

Parameter	Requirement	Measured Value
Course Completion Time	<5 minutes	TODO
Obstacle Detection Range	>2 meters	TODO
Obstacle Reaction Time	<250 ms	TODO
Remote E-stop Range	>100 ft	TODO
Navigation Accuracy	<1 meter	TODO
Battery Life	>5 minutes	TODO
Ramp Ascension	Successful	TODO

8.5 Software Testing, Bug Tracking, and Version Control

The software subteam utilizes Git and GitHub to store and version control the codebase. By utilizing branches for each new feature, those features can be isolated and independently tested before being merged into the main codebase.

TODO: Describe your CI/CD pipeline, testing framework, and bug tracking process.

8.6 Simulation

TODO: Describe any simulation environment you use (Gazebo, custom simulator, Unity, etc.).

9. Unique AutoNav and Self Drive Software

ROBOT NAME is competing in AutoNav only / is competing in both AutoNav and Self Drive.

AutoNav and Self Drive share many core features such as lane keeping and obstacle avoidance. However, Self Drive requires additional capabilities such as:

- Lane changing
- Stop sign detection and compliance
- Merging
- Parking

TODO: Describe any unique software features for each mode.

10. Performance Assessment

As of submitting this report, *ROBOT NAME* is on track for competition. Through extensive testing, the drivetrain, vision pipeline, and navigation systems have been validated.

TODO: Summarize overall readiness for competition. Be honest about what is complete and what remains.